Garrett Comes

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Thermodynamic-Computing Research — Independent, 2026

Energy-Limited Library Coverage — When Thermodynamic Hardware Beats GPUs for Discrete Design **Jun 2026 – present**

- Authored a preprint measuring when a TSU beats GPUs on the energy cost of protein and drug design, reframing the question from per-sample accuracy to sample *volume, energy, latency, and coverage*, and identifying AI-for-science Potts models, the protein-design coevolutionary landscapes, as the regime where block-parallel Gibbs sampling on sparse contact graphs is maximally differentiated.
- Quantified a coverage frontier across **59 iid $q{=}20$ Potts landscapes**, where a quench-free 80-sweep sampler reaches 99% motif coverage in bounded trial budgets and the sampling stage costs about $10^6\times$ less energy than a GPU for equal coverage, then bounded the honest end-to-end ratio, which stays throttled by shared digital readout and matching.
- Stress-tested on real RAS-superfamily coevolutionary physics and showed a localized-interface binder formulation restores full target separation and coverage. Subjected every headline claim to an adversarial review gate that scoped an initial $10^6\times$ energy claim into a precise conditional one, reproducing bit-for-bit with 9/9 regression tests passing.
- Deep working knowledge of THRML internals, including block coloring, factor shapes, sampler scheduling, and JAX/GPU simulation of TSU-style hardware.

gthrml — 16-Node Thermodynamic Sampling Unit Hardware | *Python, LTspice, THRML/JAX, KiCad* **2026**

- Designing an actual 16-node p-bit thermodynamic sampling unit as a mixed-signal board, with a boost and high-voltage noise rail, Zener avalanche noise sources, an AD8648 gain stage, and a TLV3214 comparator stage that turns physical noise into stochastic bits.
- Runs a THRML 16-node Ising computation, maps the resulting node marginals onto calibrated DAC comparator thresholds, and validates the full board in behavioral LTspice simulation.

Masked-Diffusion LM $\times$ THRML Block-Gibbs Joint Decoding | *JAX, THRML, PyTorch, flash-attn* **Apr – May 2026**

- Hybridized a pretrained masked-diffusion language model with THRML block-Gibbs joint sampling over a factor graph, where joint decoding beat independent argmax by **7–15 pp** on a held-out test split, with hard-equality factor maps dominating the alternatives.
- Probed the "no classical escape" hypothesis and retired it honestly once warm-started AIS/SMC crossed every tested energy barrier, which closed the per-sample-accuracy axis and redirected the work toward the coverage-per-energy result above.

Systems & Simulation Projects

Valurile — GPU Lattice-Boltzmann CFD Solver | *Rust, CUDA* **2024 – present**

- Multi-crate LBM solver with Esoteric Pull in-place streaming that roughly halves memory traffic per timestep, static mesh refinement, Smagorinsky–Lilly LES, SRT/TRT/MRT operators, and wgpu visualization, with GPU-side compression planned to cut PCIe transfer overhead.

repx — eBPF Reproducible-Build Attestation | *Rust, eBPF* **2026**

- Kernel-level tracing of process and file events canonicalized into Merkle-rooted attestations with content-addressed dataflow DAG provenance, verifying that re-runs reproduce identical covered operations.
- Userspace Rust CLI over Aya eBPF probes emits a SLSA-style JSON predicate, and a failed re-run walks the two trees to report which nodes diverged rather than a bare pass or fail.

SCRAP + fem_rust_ae | *FEniCSx, Rust/WASM* **2025 – 2026**

- Senior design with an AIAA SciTech 2027 extended abstract in preparation, a coupled EM–thermal quench simulation of a YBCO superconducting coil for orbital-debris detumbling, and separately a from-scratch Rust FEM solver with a live in-browser WASM demo.

Experience

Lockheed Martin Space — Propulsion Engineering Intern, Civil & Commercial Space **Summers 2022 – 2025**

Flight programs incl. Dragonfly and Mars Sample Return *Waterton Campus, Denver, CO*

- Designed a piston-tank propellant gauging system in which hermetically sealed internal magnets are read by radiation-hardened Hall-effect sensors, sized with optimization-driven magnetic simulation. Built propellant slosh and FSI models in Ansys and pure slosh CFD in FLOW-3D with test-rig correlation, and contributed to a hydrazine-replacement green-propellant trade study.

Education

Georgia Institute of Technology **Jul 2022 – Aug 2026**

B.S. Aerospace Engineering, Concentration in Propulsion *Atlanta, GA*

- Coursework in Computational Fluid Dynamics, Finite Element Analysis, Control System Design, Jet & Rocket Propulsion, and Aerothermodynamics.

Skills

Core: JAX, PyTorch, CUDA/Triton, Rust, Python, C++, probabilistic inference incl. Gibbs, AIS, SMC, and parallel tempering, engineering optimization

Tooling: Linux/NixOS, Git, MPI/PETSc/FEniCSx, agentic AI development with Claude Code, MCP servers, and custom agents